

Age norms and compulsory schooling: United States, 1850-1900

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Conference Abstract (~350 Words)

Behaviours are routinely assessed against age-related norms, which shape expectations of future aptitudes, capabilities, and welfare. Academic achievement, measured by age-standardized testing, is one such benchmark. Yet historically, age was a poor predictor of academic performance because schools were not age graded. Lasse (2013) argues that compulsory schooling policies, by defining a maximum school-entry age, strengthened the empirical association between age and academic performance and created strong, generalized age norms that shaped subsequent education policy. We test this hypothesis in the nineteenth-century United States.

Using staggered difference-in-difference (Deaton et al. 2024; Wooldridge 2025) and complete-count census microdata, we first demonstrate that compulsory schooling laws decreased attendance in the age range 4-7 and had little effect at other ages. This suggests that the laws delayed school entry. We develop an estimator to quantify this shift, showing that compulsory schooling delayed school entry by around eight months. We explore heterogeneity in the laws, including newly digitised clauses on school-entry minima, to measure how closely behavioural changes followed statutory prohibitions.

Next, we consider the substitution effect of compulsory schooling. If households' preferences over the length of school careers were unchanged by the policy, later mandatory school-leaving ages may have induced a reallocation of school careers towards later childhood. Under a pure substitution effect, delayed leaving should induce equal delays in entry. We test this by measuring the effect of compulsory education on school-leaving age.

Neither explanation wholly explains observed changes in school-entry age. This suggests compulsory schooling shaped behavior through channels beyond direct enforcement or substitution. Indeed, recent work suggests that law may affect outcomes by expressing social preferences (Bénabou & Tirole 2026; Leibovitch & Teichman 2025).

We argue that this historical episode illustrates the “expressive function of law” (Sunstein 1996). School-entry age was previously ill-defined, but compulsory schooling laws coordinated school-entry decisions. Laws may change norms even when their prohibitions are not binding.

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Extended Abstract (~1,000 words)

When should children learn to talk? When is it too late to change careers? At what age should young adults start a family? Age norms link biological age to specific life events, and deviations from these norms are associated with social costs (Settersten 2003). In particular, delay and precocity in relation to age norms can shape expectations of future aptitudes, achievements, and welfare, and these expectations can be self-reinforcing. Schools provide a rich institutional setting in which to observe age norms. For example, standardized testing in age-graded classrooms and parents' strategic manipulation of school entry around cutoffs (i.e., 'redshirting') both reflect and reinforce age-based expectations. Yet because schools were historically not age-graded, age was a poor predictor of academic performance in the past and could not have supported such expectations. Lasseonde (2013) argues that compulsory schooling laws, by coordinating school-entry decisions, linked age to academic development and created the age norms against which school policy would evaluate children for generations. We test this hypothesis in the nineteenth-century United States.

Using complete-count census microdata from IPUMS (Ruggles et al. 2024) spanning 1850–1900, we observe school attendance by age at the individual level across the full rollout of compulsory schooling laws outside the south. We use Wooldridge's (2025) staggered difference-in-difference methodology, which addresses well-known problems with TWFE estimators in staggered settings (e.g. Roth, Sant'Anna, Bilinski & Poe 2023), applied to repeated cross sections following Deb et al. (2024).

First, we demonstrate the effect of compulsory schooling on school-entry age. We focus on white U.S.-born children in northern states to minimize confounding from migration (Bandiera, Mohnen, Rasul & Viarengo 2019) and violations of the parallel trends assumption due to Reconstruction and Jim Crow policies in the South (Jones & Schmick 2025). Compulsory schooling decreased enrolment in the age range 4-7 but had little effect elsewhere. This null result at older ages is inconsistent with existing estimates using the TWFE estimator, which tend to find relatively modest positive effects (e.g. Clay et al. 2021), but is consistent with an older qualitative literature suggesting compulsory schooling laws were weakly enforced (Tyack, James, and Benavot 1987; Landes and Solmon 1972; Katz 1976). The negative result at younger ages, however, is novel and suggests that the laws delayed U.S-born children's school entry.

We explore several possible explanations. First, we test whether compulsory schooling laws were paired with other policies affecting school-entry age. Some states introduced school-entry minima to exclude young children from elementary education. We digitise these provisions to test whether our measured effect reflects such statutory prohibitions. The evidence suggests these states did not drive increases in school-entry age.

Legal restrictions may induce individuals to substitute alternative behaviors not proscribed by the law (Stigler 1970). If families' preferences over the length of school careers were unchanged by compulsory schooling, those compelled to keep their children enrolled for longer may have delayed entry by an equivalent amount. The evidence suggests this was not the case. Before compulsory schooling laws, enrolment rates plateaued between ages 9 and 13. Compulsory schooling laws tended to adopt very weakly binding entry and exit ages that followed existing enrolment patterns. We find no evidence of an increase in enrolment beyond these ages that would be consistent with later school exit. We test formally whether compulsory schooling caused longer school careers and find no evidence of this. School careers do not appear to have been extended so there is no empirical basis for a theoretical substitution effect.

In most cases, compulsory schooling laws did not require parents to delay their children's entry into school, nor was there a substitution effect, yet entry ages rose by approximately eight months following the policy. Rather, we propose that compulsory schooling provides an example of the “expressive function of law” (Sunstein 1996; Bénabou & Tirole 2026). By codifying expectations around school entry, which were previously ill-defined, legislators may have expressed social preferences about how education should be scheduled, and parents adopted these same preferences (Lassonde 2013).

This finding contributes to three debates. First, we document an unexpected consequence of compulsory schooling laws. While the literature has tended to focus on deferred workforce entry and returns to an additional year of adolescent human capital accumulation (e.g. Clay et al. 2021), our findings suggest that delayed school entry may confound these estimates in ways that have not been accounted for. Notably, studies of compulsory schooling that rely on retrospectively reported grade completion data (e.g. Puerta 2011; Clay et al. 2021) may be subject to measurement error correlated with treatment if compulsory schooling redefined the expected relationship between age and academic “grade” itself. Second, although cross-group differences in age norms are well documented (Mortimer et al. 2005), their origins have received less attention. Our study contributes to a small literature on how institutions create age norms. Prior work has shown, for instance, how Bismarck's pension scheme coordinated expectations around retirement age (Young 2015); we show that compulsory schooling laws played an analogous role for school entry. Third, and more broadly, our findings provide rare evidence, historical or otherwise, for the theory that law can shift behavior by coordinating social expectations rather than through direct enforcement or material incentives (Lane et al. 2023).

Working Draft

1. Introduction

The age at which children should begin school received serious attention only belatedly in the development of education policy. When the U.S. Commissioner of Education for the first time studied school-entry age in his report of 1887-88, he found no consensus. He cited the opinion of Mr. Calvin Peterson, city superintendent for Brooklyn, N.Y., who believed earlier school entry accelerated academic progress, and of Mr. William H. Maxwell, also city superintendent for Brooklyn, who argued “the average child is injured both physically and mentally by being put too early to the largely mechanical work of learning to read, write, and cipher” (Commissioner of Education, pp. 210-11). That two contemporaneous administrators in the same city held such opposing views was unremarkable. Since the colonial era, and still in much of the country, American children had entered school at no standard age, arriving anywhere from early childhood to adolescence (Pollock 1983; Lassonde 2013).

Compulsory schooling laws forced the issue. These policies aimed to set a floor on the number of years U.S. children had to spend in school and to limit child labour by setting a minimum school-leaving age. This in turn required a maximum school-entry age because the minimum-years requirement could not otherwise be enforced. Before the law, the modal child was sent to school at age six; the modal compulsory schooling statute required children to begin school by age eight, but parents were free to enrol their children at younger ages (Lassonde 2013). Because of the wide variance in school-entry age, the law would nonetheless have been binding for some families. If the law acted primarily to penalise late school-entry, it should have driven down mean school-entry age. We find no evidence of this.

Using a staggered difference-in-differences framework (Wooldridge 2025; Deb et al. 2024), we instead observe that compulsory schooling raised school-entry age by approximately

eight months. We situate this result in a literature on the signalling and coordinating function of the law (Sunstein 1996; McAdams 2000; Bénabou & Tirole 2026). Compulsory schooling laws, which were widely publicised in newspapers, both announced an entry age and raised the salience of the issue, providing a focal point around which previously dispersed expectations could coordinate. We explore a number of alternative explanations. Substitution between earlier and later schooling, bundled policy treatments at the state level, and administrative responses at the local level are all inconsistent with the evidence. Households appear to have voluntarily changed their behaviour in a manner the laws suggested but did not require. Laws changed school-entry age expectations.

This finding contributes to three literatures. First, we document an unexpected consequence of compulsory schooling laws. While the literature has tended to focus on deferred work-force entry and returns to an additional year of adolescent human capital accumulation (e.g. Clay et al. 2021), our findings suggest that delayed school entry may confound these estimates in unrecognised ways. Notably, studies of compulsory schooling that rely on retrospectively reported grade completion data (e.g. Puerta 2011; Clay et al. 2021) may be subject to measurement error correlated with treatment if compulsory schooling redefined the expected relationship between age and academic “grade” itself. Second, although cross-group differences in age norms are well documented (Mortimer et al. 2005), their origins have received less attention. Our study contributes to a small literature on how institutions create age norms. Prior work has shown, for instance, how Bismarck’s pension scheme coordinated expectations around retirement age (Young 2015); we show that compulsory schooling laws played an analogous role for school entry. Third, and more broadly, our findings provide rare evidence for non-coercive causal channels in

law (Lane et al. 2023), specifically the theory that law can shift behaviour by coordinating social expectations.

Section 2 gives the historical context, describing pre-compulsory treatment age and the structure of compulsory education laws. Section 3 describes the FLEX difference-in-differences methodology, our summary estimator for mean school-entry age, and the data we use to estimate the effect of compulsory education on mean school-entry age. Section 4 presents our main estimates of that effect. We find that compulsory education raises school entry age by around 8 months. Section 5 explores a variety of hypotheses to explain this finding. By elimination, we argue the explanation most consistent with the evidence is that compulsory education created a focal school-entry age, and section 6 presents preliminary positive evidence in support of this theory. Section 7 concludes.

2. Historical Setting

School-entry age varied widely in the past. Examples of school-entry age from seventeenth-century diaries range from 3 to 12 (Pollock 1983), and around a third of literate indentured servants in colonial America became literate after age 15 (Galenson 1981). Wide variation in school-entry age is also evident in the mid-nineteenth century census. Take the age distribution of students in Ohio in 1850, for example, depicted in figure 1. It shows that the youngest children entered school at age three, and the proportion enrolled increased with age as new children enrolled, before reaching a maximum around age ten. Roughly 95% of children entered school between the ages of 3 and 9.

[Figure 1 about here.]

In general, chronological age had relatively little salience in education policy before the close of the nineteenth century. Although empirical studies of child development began to emerge in the late nineteenth century, the discipline of developmental psychology was not yet sufficiently developed to inform pedagogical theory (Henderson 2024). For example, only 17% of European school manuals in the nineteenth century advocated organising classes by age (Caruso 2023). Irregular school-entry age carried through to create irregular age-distributions in higher classes. For example, in Boston's Grammar and High schools, the ages in the highest class in 1881 ranged from 13 to over 18 years old (Boston School Committee, 1881). Moreover, political activists and educational reformers were preoccupied with raising school-leaving age to the neglect of school-entry age. For example, Edward Bellamy, an early American socialist, argued in an editorial published in 1889 that raising the school-leaving age would improve the quality of civic life and contribute to "the evolution of a better social order", completely ignoring the possibility that lowering the compulsory school-entry age, which was then eight, could in principle have the same effect (*Workman's Advocate*, June 29, 1889). When the U.K. adopted legislation to introduce compulsory education in 1876, the clauses concerning school-entry age passed without debate (Henderson 2024).

[Figure 2 about here.]

The contrast with contemporary education policy is striking. Looking again at Ohio, for example, where the compulsory school-entry age is currently six, figure 2 demonstrates near-universal compliance at the compulsory age. Pre-compulsory enrolment occurs within a narrow range, with roughly 95% of children enrolling between the ages of 3 and 6. Additionally, school-entry age is highly salient today, with some parents strategically manipulating school-entry around calendar-date cutoffs to secure advantages for their children (e.g. Lenard and Peña 2018).

School-entry age policy has also received considerable attention in recent years (e.g. Bedard & Dhuey 2011). When did these changes occur?

While the contemporary literature on school-entry in developing economies emphasises how household deficits cause delayed entry relative to institutional age standards (e.g. Glewwe & Jacoby 1995; Gansaonré et al. 2022), the historical literature emphasises the laxness of institutional standards themselves. In particular, historians emphasise how institutions created more uniform experiences of childhood largely for reasons of organisational expediency as they grew in size and complexity (Graff 1995; Lassonde 2013). These narratives point to a coordination problem. Because age historically had such little salience in education, age was a poor predictor of academic performance. It was thus difficult to form expectations about how children's abilities should develop over time (e.g., see appendix 1). For educators, this meant it was difficult to design age-appropriate curricula, and for parents it meant there was little incentive to target specific school-entry ages.

Lassonde (2013) argues that an unintended consequence of compulsory education was to resolve this coordination problem. Compulsory education laws were adopted for a variety of reasons, but their immediate goals were to reduce child labour by keeping adolescents in school and to establish a floor on years of school attendance. Establishing a compulsory school-entry age followed almost as a logical necessity. Once established, however, according to Lassonde, "Because it was mandated that they begin school at the same age ... as each cohort moved through graded schools, their ages became, de facto, more uniform over time." This coordinated expectations, which were then adopted in subsequent educational standards (Lassonde 2013). We test this hypothesis in the nineteenth-century United States.

The typical compulsory schooling law specified a maximum school-entry age, a minimum school-leaving age, the minimum length of the school year, and conditions on child labour (Clay et al. 2021). Taking Ohio as an example again, its 1877 law required:

That every parent, guardian, or other person in the state of Ohio, having charge or control of any child or children between the ages of eight and fourteen years, shall be required to send such child or children to a common school for at least twelve weeks in each school year ... at least six weeks of which shall be consecutive.

On child labour, it required that:

No manufacturer, owners of mills or mines, agent, overseer, contractor [etc.] ... shall ... employ any child under fourteen years of age during the established school hours of the locality, unless such child shall have attended some common or private school for the term of at least twelve weeks during the school year next preceding the commencement of such employment.

The law subjected non-complying parents or guardians to a fine of 2 dollars for the first offense and a fine of 5 to 10 dollars for subsequent offenses. Note that the law did not define a statutory entry age. Parents were free to enroll their children before age eight, which acted only as a maximum. Whether compulsory schooling laws nevertheless shifted school-entry age, and through what mechanism, is the question we now address.

3. Data & Methodology

To estimate the effect of compulsory schooling on school-entry age, we apply the FLEX difference-in-differences estimator (Deb et al. 2024) to complete-count US census data digitised by

IPUMS for the years 1850-1900, excluding the 1890 census, which was destroyed in a fire.³ Because we are interested in families at a particular time in their life cycle, i.e., when their young children are entering school, the decade interval between censuses means that the set of families observed at this life-cycle stage differs from one census to the next. We therefore treat the census as a repeated cross-sectional dataset, assuming that each census year represents a new random draw from an underlying state superpopulation. While census-linking is possible, it is not methodologically necessary for our question and may introduce selection bias associated with linkage. FLEX extends Wooldridge's (2025) recent staggered DiD framework to repeated cross-sections. For our purposes, the main advantage of FLEX over alternatives in this literature (e.g., Callaway and Sant'Anna 2021; Borusyak et al. 2024) is that it reframes what are otherwise multi-step procedures as a single OLS regression with a transparent specification and identification framework.

Identification relies on four assumptions: the stable unit treatment value assumption (SUTVA), which requires that potential outcomes in a state depend only on the treatment status of that state; the no bad controls assumption, which requires that covariates do not change with treatment; the conditional no anticipation assumption, which requires that individuals in treated states do not change their behaviour in advance of treatment; and the conditional parallel trends assumption, which requires that counterfactual outcomes differ only in levels. Of these, we treat parallel trends as the most consequential assumption, and our sample restrictions are designed primarily to make it more credible.

We restrict our sample to white U.S.-born children in northern states to minimize confounding from migration and violations of the parallel trends assumption due to Reconstruction and

³ Note that for the working version of the paper we use a random 1% sample to speed up computation time. Because standard errors on our primary estimates are clustered at the group level (states), any improvements in precision from using the full dataset is likely to be modest.

Jim Crow policies in the South (Jones & Schmick 2025). Bandiera, Mohnen, Rasul & Viarengo (2019) show that the rate of compulsory schooling adoption across U.S. states correlates negatively with the exposure of immigrant populations to compulsory schooling in their sending countries. If our hypothesis is correct—that compulsory schooling affects school-entry age—then this correlation implies that immigrant families' school-entry trajectories diverge from those of native-born families in ways that depend on state-level treatment timing, violating parallel trends. We therefore exclude non-U.S.-born children from our sample. We note also that the American Civil War (1861-65) and its political aftermath put southern states on an educational trajectory distinct from the rest of the country. Notably, Reconstruction era policies expanded Black educational opportunities, but Jim Crow retrenchment, beginning in the 1880s and 1890s, reduced state education expenditure for both Black and white children in the South thereafter (Jones & Schmick 2025). Further, because all former confederate states adopted compulsory schooling between 1910-1920, there is no within-region variation across census years to exploit for within-group identification. Because Black migration from the South accelerated after the Civil War, the population of Black residents in the North changed qualitatively over our study window, raising further parallel-trends concerns for this group.

Prior work suggests that compulsory schooling policies lagged earlier increases in school supply (Goldin 1999). This raises a further parallel-trends concern. If untreated states are simply earlier on the same developmental path, they would continue expanding school supply over the study period while treated states, further along, would not. To minimise this concern, we focus on urban districts,⁴ which tended to develop more rapidly than rural districts so that school

⁴ IPUMS applies a uniform definition of urban/rural status to all pre-1950 censuses, based on the methodology used in the 1930 census: “Although still, for the most part, comprised of cities and incorporated places of 2,500 inhabitants or more, the 1930 census returns also considered townships and other political subdivisions (not incorporated as

supply was likely adequate even in untreated states.⁵ Further, section 5 draws on data available only for urban districts, and we maintain the urban restriction in our main analysis for consistency. For comparability with previous work (e.g. Clay et al. 2021), we also restrict our sample to males.⁶

The most likely violations of the no anticipation and SUTVA assumptions would arise from behavioural diffusion across state lines, where families might learn about changes in school policy in neighbouring states. As discussed, we think that families learned about compulsory schooling via local newspapers, which would be unlikely to circulate beyond their immediate communities, and not via national media. In such a situation, however, the most likely analytical consequence would be attenuation of treatment effects because the control group would behave more like the treated group. For these reasons, we generally consider treatment effects estimated on the initial wave to be the most robust, as the first-treated cohort had no already-treated neighbours to learn from at the time of their own treatment.

[Table 1 about here.]

Our main outcome is school enrolment, measured by the census question asking whether children had enrolled in the past year at the time of the census, from digitised census manuscripts (Ruggles et al. 2025). We take the year a state first introduced compulsory schooling from the supplementary material published by Clay et al. (2021). We exclude Vermont because it first received treatment in 1867 and was the only state treated at the time of the 1870 census. We also exclude Arizona, Wyoming, and Oklahoma because these territories introduced compulsory

municipalities) having a total population of 10,000 or more, and a population density of 1,000 or more per square mile, as urban” (IPUMS 2026).

⁵ A long historiography has argued that the U.S. is the sole historical example of rural school supply exceeding urban, but this is based on a misreading of the evidence; see Lachanski (2026).

⁶ We intend to analyse heterogeneity between boys and girls in later work.

schooling in the same decade they became organised states, leaving no untreated periods in the data.

We further assemble a summary measure of the effect on mean school-entry age by recasting the student age distribution as a survival curve. First, assume that children enrolling in school below age a^* do not leave school before age a^* . That is, below age a^* , school enrolment is an absorbing state. Then the share of children enrolled at any age $a \leq a^*$, must represent the cumulative share of children who enrolled before age a . In other words, the cumulative distribution function (CDF) for school-entry age is

$F(a) = \Pr(\text{enrolled} a) \text{ if } a \leq a^*.$	$(*)$
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After age a^* , children may leave school, and the share enrolled will fall with age once net departures exceed new enrolments. Age a^* is therefore naturally interpreted as the mode of the student age distribution. If the number of students who enrol after age a^* is negligible, the proportion of students enrolled at age a^* is approximately equal to the proportion of children who ever enroll in school.

Under these assumptions, we recast the CDF as a survival function and calculate the expected school-entry age, $E(e)$, via the tail-sum property,

$E(e) = \sum_{a=0}^{a^*} [1 - F(a)] = \sum_{a=0}^{a^*} S(a).$	$(*)$
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Conditional on ever enrolling in school, this becomes

$E(e \mid \text{ever enrolled}) = \frac{1}{F(a^*)} \sum_{a=0}^{a^*} S(a).$	$(*)$
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Both the survival function and the proportion enrolled at the mode are recoverable from the linear probability model described above. Note that the difference in mean enrollment age across treated and untreated groups is the effect of treatment on mean enrolment age

$\begin{aligned}\Delta E(e \mid \text{ever enrolled}) &= E_T(e \mid \text{ever enrolled}) - E_C(e \mid \text{ever enrolled}) \\ &= \frac{1}{F(a^*)} \sum_{a=0}^{a^*} S_T(a) - \frac{1}{F(a^*)} \sum_{a=0}^{a^*} S_C(a) \\ &= -\frac{1}{F(a^*)} \sum_{a=0}^{a^*} [F_T(a) - F_C(a)] = -\frac{1}{F(a^*)} \sum_{a=0}^{a^*} \text{TE}(a).\end{aligned}$	$(*)$
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In words, the effect on mean school-entry age is the negative of the sum of all the age-specific treatment effects up to the mode of the student age distribution, divided by the proportion enrolled at the mode. We further assume that the ever-enrolled proportion does not change with treatment. If the proportion ever-enrolled does change, we would need to account for their effect on the conditional school-entry age. In principle, this effect could be bounded by assuming extreme upper and lower entry-ages for new enrollees in a method analogous to Lee bounds (Lee 2009; Rathnayake et al. 2024). In practice, our empirical results do not suggest compulsion increased the proportion ever enrolled, or at least its effect was very small.

[Figure 3 about here.]

4. Results

Figure 3 plots the average treatment effect on the treated (in blue) at each age from 3 to 16. The treatment effect is averaged across all treated states observed in post-treatment periods 1880 and 1900, weighted by the number of treated children in each state-year cell. Deviations below (above) 0 indicate that compulsory schooling reduced (increased) the age-specific enrolment rate

by the amount indicated on the y axis. The age distribution of all not-yet-treated students across all pre-treatment periods is rendered in the background as a heuristic reference point. Note that very few three- and four-year-olds enrolled in school in the pre-treatment group, but the enrolment rate jumped from 6% to 23% between age five and six. The pre-treatment enrolment rate increased most steeply, from 23% to 55%, between age six and seven, indicating that the modal pre-treatment school entry age lay in this interval. The pre-treatment enrolment rate continued to increase, tapering off gradually before reaching a maximum of 88% at age ten. It is over this same age range, when new children were entering school and causing enrolment rates to rise with age, that the effect of compulsory education was greatest.

Compulsory education had a negative effect on enrollment rates between the ages of four and 10, with the sharpest decrease of 13 percentage points occurring at six, the modal entry age. The estimated treatment effect by age remained negative until age 11, suggesting that the whole school-entry age distribution shifted to the right in response to compulsory education. As table 2 notes, our summary measure indicates that compulsory education laws raised school-entry age by roughly eight months in 1880 and by roughly four months in 1900. The smaller treatment effect in 1900 is accompanied by a rising mean school-entry age in untreated states from 6.49 to 6.86 years (Table 2), almost exactly offsetting the apparent decline. This suggests that by 1900 it is more appropriate to view untreated states as “catching up” with treated states than to interpret the smaller estimate as a weakening of compulsory schooling's effect, pointing to a behavioural diffusion mechanism affecting untreated states over time.

For interested readers, appendix Tables 1 and 2 reproduce the full regression table using a “lags and leads” specification. Interactions between treated states and pre-treatment time periods provide a test of the parallel trends assumption at each age. With 39 tests for violations of

parallel trends, some inevitably reject the hypothesis of no pre-trend. Where violations of pre-trends exist, they suggest that states adopting compulsory education earlier had higher enrolment rates in the 1850s and 1860s broadly across the whole age distribution, in line with the earlier literature (Goldin 1999), but there is no sign of an age-specific pattern.

[Table 2 about here.]

5. Alternative explanations

Our findings present a puzzle. Compulsory education laws did not formally apply to the ages at which their effect was greatest. In 71.9% of the laws we study, the statutory school-entry age was eight (see Table 3), operating as a maximum. In the vast majority of states, parents were free to enrol their children at younger ages if they wished. Indeed, as we show above, most parents did. Why, then, were the laws most effective at changing the behaviour of children below the statutory entry age, on whom their legal provisions had no obvious bearing?

We first consider the hypothesis that compulsory education laws were bundled with policies prohibiting young children from entering school (Goldin & Katz 2008). Using the HeinOnline legal database, we check the text of all the statutes catalogued in Clay et al. (2021) for clauses restricting enrolment at young ages. We identify only two such clauses among the initial compulsory schooling laws adopted across states. These are Wyoming (1873 Wyo. Sess. Laws 247) and Montana (1883 Mont. Laws 56). Arizona introduced similar clauses in later education statutes, but there was no such clause in the initial law. The force of such clauses would have been to lower school enrolment rates below the statutory school-entry minimum, followed by a sharp recovery as all the children who would have enrolled earlier delayed entry to comply with the statute. This is not what we observe in the data; rather, compulsory schooling laws broadly

shifted the whole school-entry age distribution to the right. Moreover, because states that enacted school-entry minima were sparsely populated during the period of our study, it is unlikely that they account for much of the average treatment effect, which is weighted by population.⁷

[Table 3 about here.]

If state-level policy does not explain the puzzle, it is nonetheless possible that local administrators introduced regulatory changes in conjunction with compulsory education. In particular, qualitative studies suggest school districts began to introduce age-grading towards the end of the nineteenth century (Vinovskis et al. 1995). School districts had already begun to organise the curriculum into hierarchical grades taught by teachers with different qualifications, but students were sorted into grade according to ability. Age-grading was an attempt to make children's age and ability correspond, and it required that children enter school at roughly the same age. Districts were able to adopt this practice without state-level direction, although they may have taken a cue from the age provisions in compulsory education laws.

Data to explore this question are scarce, but some well-organised school districts produced high-quality annual reports with information about the distribution of students by grade at the time compulsory education was introduced.⁸ Note that well-organised urban school districts are exactly where one would expect to observe any age-grading response if one existed. Grade distributions are a useful proxy for age-grading because they indicate roughly how long children remained in each grade. For example, in Cleveland in 1874, the difference in mean age between children in the first and second grades was 1.9 years, and there were 3,588 second-grade children

⁷ Indeed, these territories were not sufficiently populated at the time of the early federal censuses to have been included in the count. They are therefore missing from our full lags and leads model in Appendix Tables 1 and 2 because the data is not available to estimate leading pre-trends.

⁸ These are Baltimore, Boston, Brooklyn, Cleveland, D.C., New York, Philadelphia, Providence, San Francisco, and St. Louis. Relevant data for the state of Maryland also exist.

to 6,263 first-grade children (roughly 1:2); in contrast, the age difference between third- and second-grade children was only 1.3 years, and there were 3,109 third-grade children to 3,588 second-grade children (Cleveland Board of Education 1874, table vii). In particular, what was expected for promotion from the initial grade to the next highest grade was commonly beyond the abilities of the youngest children, creating a pronounced bottleneck at the initial primary grade.

This point is made in the Boston School District report of 1874:

The proportion in the sixth [i.e., initial, lowest] class is larger than it should be. Many children from well-to-do families, do not now enter school until they are six or seven years old. Such children should not be required to remain a half a year with the younger children in the sixth class. Still it is to be expected that, as a rule, even under the best management, the aggregate number of pupils in the sixth class will exceed that of any of the classes above it (Boston School Committee 1874).

We anticipate that age-grading would relax this bottleneck. If the content of the initial grade was adjusted to more closely correspond with the capabilities of young students, more would advance to the second grade, and the promotion rate would rise toward 100%.

We measure the promotion rate, p , by

$p = \frac{g_{2,t}}{g_{1,t-1}},$	(1)
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where $g_{i,t}$ is the number of students enrolled in grade i at time t . This should approximately capture the annual promotion rate of students from the initial grade. Using the previous year's first-grade enrolment as a denominator accounts for annual growth in school enrolment.⁹ Note that we apply linear interpolation to missing years of initial grade enrolment, e.g., where school districts report enrolment statistics every two years.

[Figure 4 about here.]

⁹ It is confounded by second-grade retention, but this appears to have been much less severe than first grade retention in the data.

Figure 4 plots coefficients from an event-study style regression showing the impact of compulsory schooling laws on our measure of grade progression. Note that we are not able to use time fixed effects as in a full DiD, as enrolment data disaggregated by grade were only reported in a limited number of years.¹⁰ Our sampling strategy was to collect as much data as was available in a ± 10 -year window around the introduction of compulsory schooling in the state. We nonetheless find no evidence of a change in grade progression around the introduction of compulsory schooling laws, either in the post-treatment period or as pre-treatment anticipation. We interpret as inconsistent with age-grading at the local level.

Finally, we consider whether behavioural substitution induced by legal restrictions at older ages explains our findings. Visually, figure 3 shows that the effect of compulsory schooling at older ages was statistically indistinguishable from zero. However, we can quantify the overall effect using a parallel argument to the one used to measure mean school-entry age. That is, assuming school-entry beyond the modal age was minimal, the decline in enrolment with age can also be recast as a survival curve and the age-specific effects summed to get mean school-leaving age. The outcome of this exercise is presented in table 4. Our estimates suggest compulsory education raised school-leaving age by only four days in 1880. In 1900, the effect is larger, consistent with qualitative accounts of enforcement efforts increasing over time (Katz 1973). Still, we find that compulsory education in 1900 raised school-leaving age by only 47 days, and the null hypothesis cannot be rejected. Families could not have substituted school attendance in later childhood for attendance in early childhood because their children simply did not stay in school significantly longer.

¹⁰ The underlying regression includes city fixed effects and analytic weights proportional to the number of students enrolled.

We argue that the evidence is inconsistent with policy bundling at state and local levels and with behavioural substitution. In the next section, we argue that the law likely operated as a focal point and raised the salience of school entry, changing behaviour without coercion.

6. Discussion

Our evidence is consistent with theories of the ‘expressive function of law’ (Sunstein 1996), particularly those that emphasise the law’s power to coordinate expectations around a focal point (McAdams 2000; Bénabou & Tirole 2026). We note in section 2 how school-entry age had very little salience in the pre-treatment period. Raising the salience of a particular age may be theoretically sufficient to promote coordination on a given outcome, even when parents are indifferent as to the entry-age chosen, by simply making it stand out among the available options. Mehta et al. (1994) refer to this as “primary salience” and demonstrate its relevance in an experimental setting. Compulsory schooling laws would have raised the salience of a particular school-entry age in a number of ways.

First, the laws were widely publicised and debated. For example, after New York passed its first compulsory schooling law in 1875, the *New York Tribune* published a full summary of its contents, including the statutory age prominently in the sub-heading:

PROVISIONS OF THE BILL PASSED BY THE LAST LEGISLATURE—CHILDREN BETWEEN 8 AND 14 YEARS OF AGE MUST BE INSTRUCTED IN CERTAIN BRANCHES AT LEAST 14 WEEKS IN EVERY YEAR—TREATMENT OF HABITUAL TRUANTS—PENALTIES PROVIDED FOR VIOLATION OF THE ACT (May 16, 1874 *New York Tribune*)

News spread to smaller newspapers as well. For example, as Pennsylvania’s compulsory schooling law was in the final stages of legislative approval in 1895, the *Scranton Tribune* reprinted a wire article from the *Philadelphia Press* summarizing its main provisions: “first, the duty of

parents to send their children to school under penalty; second, the necessity of officers to compel this attendance, and third, the need of a school census” (May 15, 1895 *Scranton Tribune*). The second paragraph also prominently named the statutory school ages, but these other provisions point to further mechanisms by which salience was raised. Compulsory schooling laws created administrative procedures to determine which children fell under the provisions of the law. The truant officer and the school census provided recurrent occasions on which the salience of the statutory age was raised, although again, there was nothing to discourage parents from enrolling children below the statutory age in either New York or Pennsylvania.

In this sense of “primary salience”, compulsory education laws may have made some parents more likely to opt for later school-entry. A similar mechanism has been proposed to explain bunching on retirement ages that are more saliently featured in the design of pensions but not obviously beneficial from the point of view of maximizing lifetime utility (Young 2015; Seibold 2021). However, if coordination on school-entry age does bring benefits, then we might expect to observe “secondary salience”, i.e., outcomes that individuals expect others to find more salient become themselves more salient, which experimental evidence has demonstrated is more effective at creating coordination (Mehta et al. 1994). In other words, if parents think compulsory schooling laws are likely to affect other parents’ choice of school-entry age, they might themselves alter their chosen school-entry age to coordinate. This may lead to persistence of the new school-entry age even after the immediate publicity surrounding the law has faded.

Coordination on school-entry age allows comparisons of academic performance to be made across children, holding biological age constant. All else equal, this makes academic performance more informative about underlying ability, which is otherwise confounded by

biological age. This point was made explicitly during the development of the Stanford-Binet intelligence test:

One of the most common errors made by the teacher is to overestimate the intelligence of the over-age pupil. This is because she fails to take account of age differences and estimates intelligence on the basis of the child's school performance in the grade where he happens to be located. She tends to overlook the fact that quality of school work is no index of intelligence unless age is taken into account (Terman 1916).

Better information about children's underlying abilities has obvious benefits for high-ability children, but the benefits for low-ability children are less clear. It is not strictly necessary that conflicting interests are entirely absent for more parents to coordinate than would have done in the absence of the "expressive" law (McAdams 2000). However, even parents of low-ability children may benefit from the ability to optimally reallocate resources in response to this information (e.g. compensating advantage).

Coordination could be achieved with an age grading policy, but as we demonstrated above, this was not pursued until later and did not coincide with compulsory schooling laws. In the absence of coordination, it is difficult for any individual teacher or administrator to determine the academic achievements that could be expected of any group of children. The signal is simply too noisy—a point also made by Terman in developing the Stanford-Binet test:

In the first place, in order to judge an individual's intelligence it is necessary to have in mind some standard as to what constitutes normal intelligence. This the ordinary parent or teacher does not have. In the case of school children, for example, each pupil is judged with reference to the average intelligence of the class. But the teacher has no means of knowing whether the average for her class is above, equal to, or below that for children in general. Her standard may be too high, too low, vague, mechanical, or fragmentary. The same, of course, holds in the case of parents or any one else attempting to estimate intelligence on the basis of common observation (Terman 1916).

This points to a higher coordination problem. It may be difficult for administrators and teachers to coordinate on an age-grading system before parents have coordinated on school-entry age.

This is essentially Lassonde's (2013) argument: age-grading was downstream of compulsory education's raising the secondary salience of school-entry age.

Although our evidence is consistent with both primary and secondary salience, we tentatively propose two reasons to favour secondary salience. First, the effect persisted. If public discussion of compulsory school-entry age raised the salience of a particular age, we would expect this to be temporary as newspaper coverage of compulsory schooling faded out. Instead, we demonstrate (in Table 2) that although the estimated treatment effect declines by 1900, this is largely driven by rising school-entry age in untreated states. This is likely related to spillovers in the salience of compulsory school-entry age, as untreated states debated compulsory education and drew on examples from their neighbours. Second, although we found no short-run effect of compulsory schooling on age-grading and so reject the hypothesis that local administrative change was deliberately tied to state compulsory education laws, we do find long-run evidence of more regular enrolment patterns in states with longer histories of compulsory education.

[Figure 5 about here.]

Although grade-distribution data are rare, the federal Education Commission began to collect such statistics from 1907. They were only reported in the commissioner's reports for a few years, 1907-1909, although further statistics reportedly exist in manuscript form. Figure 5 plots the progression rate defined in equation (1) calculated from these data against the date when a state first adopted compulsory education. It shows that states that adopted compulsory education in the 1870s had progression rates roughly 10 percentage points higher than those that adopted them in the 1900s. We stress that this evidence is preliminary and suggestive, but it is consistent with secondary salience leading to coordination in the long run. We intend to do more work to disentangle primary from secondary salience.

Our findings speak to recent work on the long-run effects of compulsory schooling. While the literature has tended to focus on deferred workforce entry and returns to an additional year of adolescent human capital accumulation (e.g. Clay et al. 2021), our findings suggest that delayed school entry may confound these estimates in ways that have not been previously noted. First, changes in school entry age may change the quality of education. There may be “real” effects related to the scheduling of human capital accumulation relative to sensitive periods of neurological development (e.g. Cunha & Heckman 2007). There may also be “nominal” effects related to grade progression. In particular, studies of compulsory schooling that rely on retrospectively reported grade completion data (e.g. Puerta 2011; Clay et al. 2021) may be subject to measurement error correlated with treatment if compulsory schooling changed the relationship between age and grade progression.

Second, improvements in the quality of information about underlying ability may provide an alternative explanation for compulsory education’s labour market effects. For example, Rauscher (2016) documents that these laws improved intergenerational mobility in the long run while reducing it in the short run, attributing the short-run effect to classroom crowding. Our evidence is in tension with the crowding mechanism. We show that the laws reduced rather than increased the number of young children in classrooms. We speculate that the social mobility effect may instead operate through information. Before compulsory schooling, when children entered school at widely varying ages, teachers could not easily distinguish ability from age, and tended to misread young children as low-ability and older children as more capable (a pattern documented in the early IQ literature, see appendix). As compulsory schooling laws coordinated entry ages across families, classrooms became more age-homogeneous, and teachers' assessments of student ability became more accurate. This information-legibility mechanism would have

disproportionately benefited poor children, who in the pre-treatment period appear to have entered school earlier than their richer counterparts (see example from Boston Committee, 1874, above). Future work should examine whether the entry-age shift documented here was concentrated among poor families, as this mechanism predicts, and whether the resulting changes in teacher assessment can account for the long-run mobility patterns Rauscher documents.

7. Conclusion

This paper documents an unexpected consequence of nineteenth-century compulsory schooling laws in the United States. Using complete-count census data and a staggered difference-in-differences design, we find that compulsory schooling raised school-entry age by approximately eight months in the first post-treatment census and four months in the second. The effect operated on children below the statutory entry age, whom the laws did not legally compel, and was largest at the pre-treatment modal entry age of six. The whole school-entry age distribution shifted to the right in response to the law.

The pattern is inconsistent with the most natural mechanical explanations. State-level policies prohibiting young children from entering school were rare and concentrated in sparsely-populated states. Local administrative responses, such as age-grading in school districts, do not show up in grade-progression data from urban districts where such responses would have been most likely. Finally, life-course substitution between earlier and later schooling cannot account for the entry-age shift because compulsory schooling had no detectable effect on school-leaving age. Our results suggest instead that compulsory schooling laws operated as a focal point, coordinating previously dispersed family expectations around school entry. The laws did not compel the behaviour they appeared to produce; they made a particular entry age salient, and families

voluntarily adjusted their behaviour toward it. This finding contributes to a growing literature on the non-coercive channels through which law affects behaviour and provides historical evidence for the construction of age norms by educational institutions.

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Table 1. Year that states first introduced compulsory schooling

Vermont	1867	Oregon	1889
Michigan	1871	Colorado	1889
Connecticut	1872	Utah	1890
Wyoming	1873	New Mexico	1891
Nevada	1873	Pennsylvania	1895
California	1874	Kentucky	1896
Massachusetts	1874	Indiana	1897
Kansas	1874	West Virginia	1897
New York	1874	Iowa	1902
New Jersey	1875	Maryland	1902
Maine	1875	Missouri	1905
Arizona	1875	Delaware	1907
Ohio	1877	Oklahoma	1908
New Hampshire	1879	Tennessee	1913
Wisconsin	1879	North Carolina	1913
D.C.	1880	Texas	1916
Illinois	1883	Louisiana	1916
Rhode Island	1883	Arkansas	1917
Montana	1883	Alabama	1917
South Dakota	1883	Georgia	1917
North Dakota	1883	Virginia	1918
Minnesota	1885	Florida	1919
Washington	1886	South Carolina	1919
Idaho	1887	Mississippi	1920
Nebraska	1887		

Source: Clay et al. (2021)

Table 2. Effect of compulsory schooling on mean school-entry age by calendar year

	1880	1900
<u>Untreated</u>		
Mean age	6.49	6.86
Proportion ever enrolled	0.84	0.95
N	6,707	2,827
<u>Treated</u>		
Effect on school entry age	0.68 (0.18)	0.31 (0.08)
N	6,229	17,505

Table 3. Characteristics of initial compulsory education laws

	N (%)
<i>Had a minimum entry-age clause?</i>	
Yes	2 (7.7%)
No	24 (92.3%)
<i>What was the statutory maximum entry age?</i>	
Seven	7 (15.6%)
Eight	23 (71.9%)
Nine	1 (3.1%)
Ten	3 (9.4%)
<i>What was the statutory minimum exit age?</i>	
Thirteen	1 (3.1%)
Fourteen	22 (68.8%)
Fifteen	4 (12.5%)
Sixteen	4 (12.5%)
Eighteen	1 (3.1%)

Table 4. Effect of compulsory schooling on school-leaving age

	1880	1900
<u>Treated</u>		
Effect on school-leaving age	0.01 (0.10)	0.13 (0.11)
N	2,934	8,203

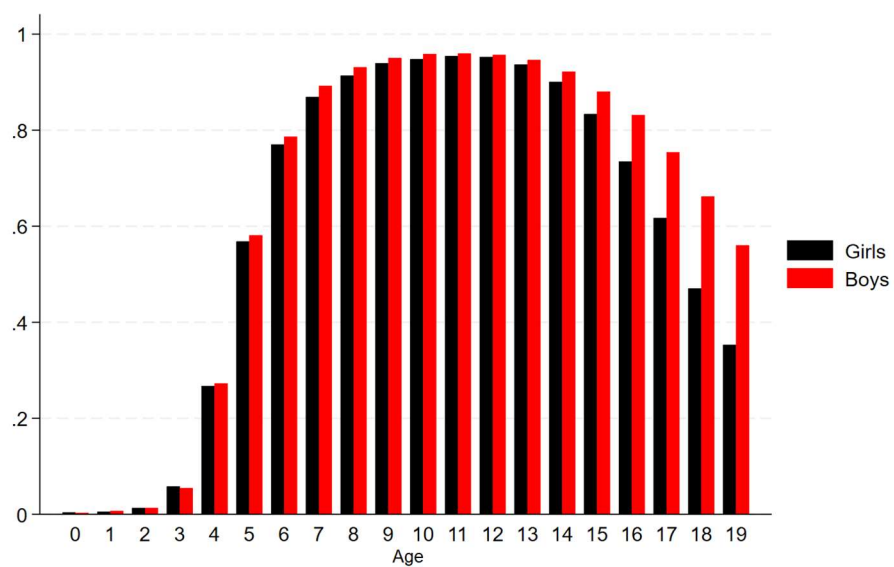


Figure 1. Age distribution of students in Ohio, boys and girls, 1850

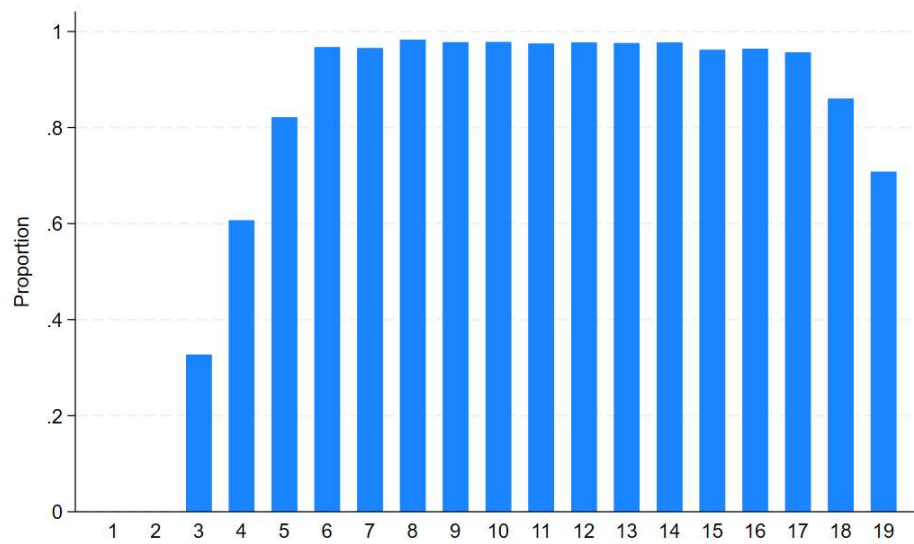


Figure 2. Age distribution of students in Ohio, 2019

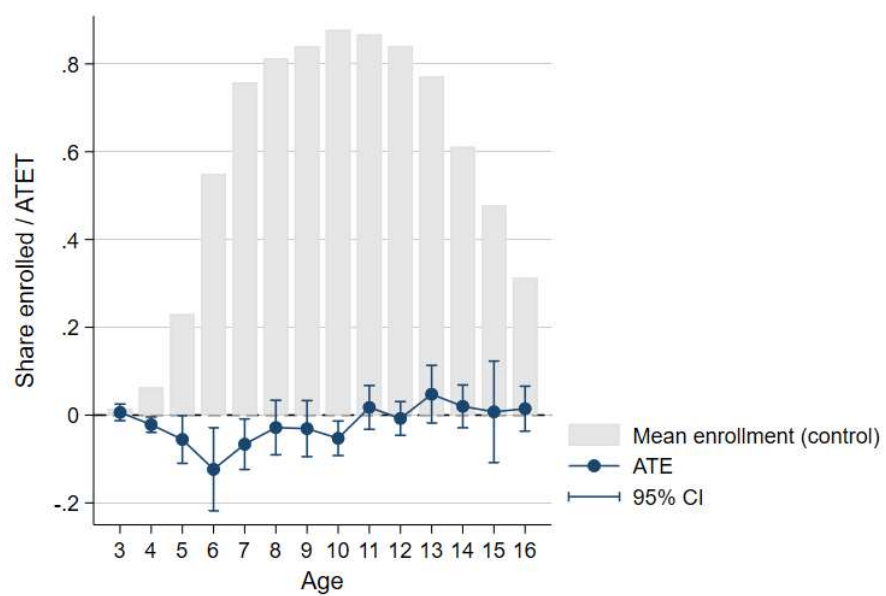


Figure 3. Effect (ATET) of compulsory schooling laws on school enrollment by age

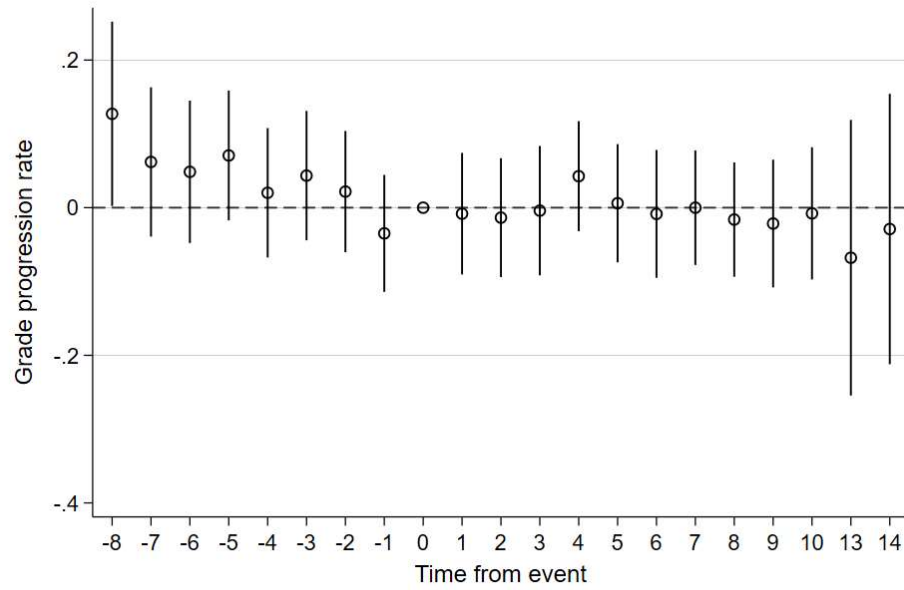
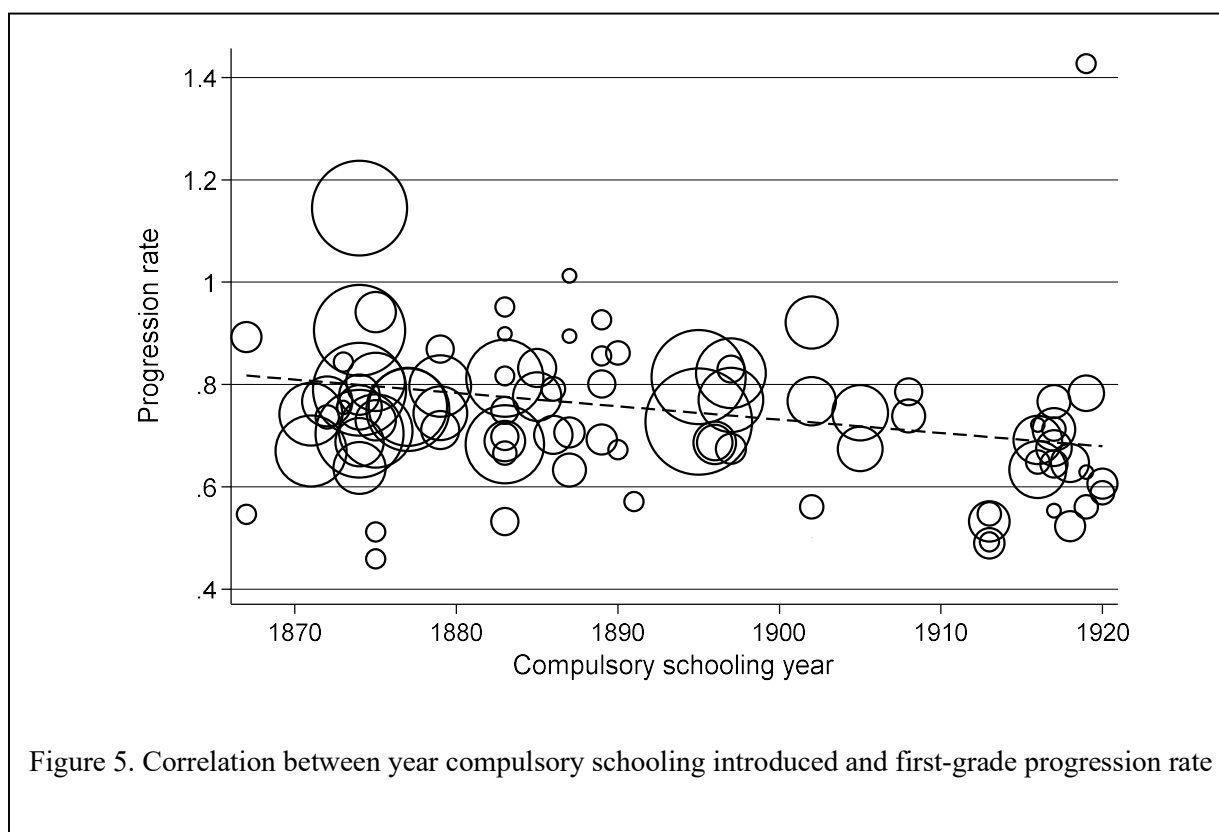


Figure 4. Response of grade progression to compulsory education laws in select urban districts



APPENDIX 1. Education as a weak signal of ability before age grading

Overestimation of ability

A. R. Girl, age 11; in low second grade. She was able to do the work of this grade, not well, but passably. The teacher's judgment as to this child's intelligence was "dull but not defective." What the teacher overlooked was the fact that she had judged the child by a 7-year standard, and that, instead of only being able to do the work of the second grade indifferently, a child of this age should have been equal to the work of the fifth grade. In reality, A. R. is definitely feeble-minded. Although she is from a home of average culture, is 11 years old, and has attended school five years, she has barely the intelligence of the average child of six years.

D. C. Boy, age 17; in fifth grade. His teacher knew that he was dull, but had not thought of him as belonging to the class of feeble-minded. She had judged this boy by the 11-year standard and had perhaps been further misled by his normal appearance and exceptionally satisfactory behavior. The Binet test quickly showed that he had a mental level of approximately 9 years. There is little probability that his comprehension will ever surpass that of the average 10-year-old.

R. A. Boy, age 17; mental age 11; sixth grade; school work "nearly average"; teacher's estimate of intelligence "average." Test plainly shows this child to be a high-grade moron, or border-liner at best. Had attended school regularly 11 years and had made 6 grades. Teacher had compared child with his 12-year-old classmates.

H. A. Boy, age 14; mental age 9-6; low fourth grade; school work "inferior"; teacher's estimate of intelligence "average." The teacher blamed the inferior quality of school work to "bad home environment." As a matter of fact, the boy's father is feeble-minded and the

normality of the mother is questionable. An older brother is in a reform school. We are perfectly safe in predicting that this boy will not complete the eighth grade even if he attends school till he is 21 years of age.

F. I. Boy, age 12-11; mental age 9-4; third grade; school work "average"; teacher's estimate of intelligence "average"; social environment "average"; health good and attendance regular. Intelligence and school success are what we should expect of an average 9-year-old.

D. A. Boy, age 12; mental age 9-2; third grade; school work "inferior"; teacher's estimate of intelligence "average." Teacher imputes inferior school work to "absence from school and lack of interest in books"; we have yet to find a child with a mental age 25 per cent below chronological age who was particularly interested in books or enthusiastic about school.

C. U. Girl, age 10; mental age 7-8; second grade; school work "average"; teacher's estimate of intelligence "average." Teacher blames adenoids and bad teeth for retardation. No doubt of child's mental deficiency.

P. I. Girl, age 8-10; mental age 6-7; has been in first grade 2½ years; school work "average"; teacher's estimate of intelligence "average." The mother and one brother of this girl are both feeble-minded.

H. O. Girl, age 7-10; mental age 5-2; first grade for 2 years; school work "inferior"; teacher's estimate of intelligence "average." The teacher nevertheless adds, "This child is not normal, but her ability to respond to drill shows that she has intelligence." It is of course true that even feeble-minded children of 5-year intelligence are able to profit a little from drill. Their weakness comes to light in their inability to perform higher types of mental activity.

Underestimation of ability

M. L. Girl, age 11-2; mental age “average adult” (16); sixth grade; school work “superior”; teacher’s estimate of intelligence “average.” Teacher credits superior school work to “unusual home advantages.” Father a college professor. The teacher considers the child accelerated in school. In reality she ought to be in the second year of high school instead of in the sixth grade.

H. A. Boy, age 11; mental age 14; sixth grade; school work “average”; teacher’s estimate of intelligence “average.” According to the supplementary information the boy is “wonderfully attentive,” “studious,” and possessed of “all-round ability.” The estimate of “average intelligence” was probably the result of comparing him with classmates who averaged about a year older.

K. R. Girl, age 6-1; mental age 8-5; second grade; school work “average”; teacher’s estimate of intelligence “superior”; social environment “average.” Is it not evident that a child from ordinary social environment, who does work of average quality in the second grade when barely 6 years of age, should be judged “very superior” rather than merely “superior” in intelligence? The intelligence quotient of this girl is 140, which is not reached by more than one child in two hundred.

S. A. Boy, age 8-10; mental age 10-9; fourth grade; school work “average”; teacher’s estimate of intelligence “average.” Teacher attributed school acceleration to “studiousness” and “delight in school work.” It would be more reasonable to infer that these traits are indications of unusually superior intelligence.

Appendix Table 1. Effect of compulsory schooling on (linear) probability of being in school, by age (ages 4-9)

	(1) OLS: Age 4	(2) OLS: Age 5	(3) OLS: Age 6	(4) OLS: Age 7	(5) OLS: Age 8	(6) OLS: Age 9
Cohort 1871-1880						
Connecticut 1850	0.221*** (0.027)	0.039 (0.036)	0.197 (0.149)	0.227* (0.095)	0.270*** (0.070)	0.231** (0.078)
Connecticut 1860	0.009 (0.012)	0.295*** (0.026)	-0.091 (0.085)	0.152 (0.080)	0.096* (0.043)	0.010 (0.115)
Connecticut 1880	-0.077*** (0.011)	-0.180*** (0.040)	-0.004 (0.126)	0.145* (0.065)	-0.162* (0.069)	0.067* (0.024)
Connecticut 1900	-0.044*** (0.009)	-0.221*** (0.032)	-0.252*** (0.056)	0.079* (0.032)	0.061 (0.039)	0.115*** (0.016)
Maine 1850	-0.014 (0.027)	0.251*** (0.036)	-0.369* (0.149)	0.081 (0.095)	0.070 (0.070)	0.273** (0.078)
Maine 1860	0.037** (0.012)	0.256*** (0.026)	-0.491*** (0.085)	-0.065 (0.080)	-0.104* (0.043)	-0.007 (0.115)
Maine 1880	0.016 (0.011)	0.118** (0.040)	-0.400** (0.126)	-0.191** (0.065)	-0.242** (0.069)	-0.328*** (0.024)
Maine 1900	-0.091*** (0.009)	-0.276*** (0.032)	-0.509*** (0.056)	0.042 (0.032)	-0.514*** (0.039)	-0.042* (0.016)
Massachusetts 1850	0.295*** (0.027)	0.125** (0.036)	-0.090 (0.149)	-0.086 (0.095)	0.072 (0.070)	0.292** (0.078)
Massachusetts 1860	0.104*** (0.012)	0.147*** (0.026)	-0.082 (0.085)	-0.018 (0.080)	-0.166** (0.043)	0.106 (0.115)
Massachusetts 1880	0.023* (0.011)	-0.070 (0.040)	-0.188 (0.126)	-0.057 (0.065)	-0.117 (0.069)	0.061* (0.024)
Massachusetts 1900	-0.010 (0.009)	-0.094** (0.032)	-0.266*** (0.056)	-0.241*** (0.032)	-0.153*** (0.039)	-0.000 (0.016)
Michigan 1850	0.000 (.)	0.623*** (0.036)	0.245 (0.149)	0.187 (0.095)	-0.812*** (0.070)	0.511*** (0.078)
Michigan 1860	-0.042** (0.012)	-0.081** (0.026)	-0.044 (0.085)	0.179* (0.080)	-0.086 (0.043)	0.311* (0.115)
Michigan 1880	-0.014 (0.011)	-0.058 (0.040)	-0.222 (0.126)	0.009 (0.065)	-0.266*** (0.069)	0.132*** (0.024)
Michigan 1900	-0.039*** (0.009)	0.038 (0.032)	-0.170** (0.056)	-0.002 (0.032)	-0.065 (0.039)	0.084*** (0.016)
New Hampshire 1850	0.236*** (0.027)	0.139** (0.036)	0.614*** (0.149)	0.414*** (0.095)	0.403*** (0.070)	0.273** (0.078)
New Hampshire 1860	0.014 (0.012)	0.345*** (0.026)	0.659*** (0.085)	0.506*** (0.080)	0.104* (0.043)	0.135 (0.115)
New Hampshire 1880	-0.007 (0.011)	-0.444*** (0.040)	0.205 (0.126)	0.227** (0.065)	0.291*** (0.069)	-0.178*** (0.024)
New Hampshire 1900	-0.022* (0.009)	-0.145*** (0.032)	0.213** (0.056)	-0.047 (0.032)	0.177*** (0.039)	-0.250*** (0.016)
New Jersey 1850	0.090** (0.027)	0.098* (0.036)	-0.181 (0.149)	0.114 (0.095)	-0.144 (0.070)	0.212* (0.078)
New Jersey 1860	0.109*** (0.012)	0.110*** (0.026)	-0.137 (0.085)	0.089 (0.080)	-0.158** (0.043)	0.034 (0.115)
New Jersey 1880	-0.029* (0.011)	-0.061 (0.040)	0.015 (0.126)	0.034 (0.065)	-0.109 (0.069)	0.045 (0.024)
New Jersey 1900	-0.012 (0.009)	0.027 (0.032)	-0.122* (0.056)	0.032 (0.032)	-0.078 (0.039)	0.012 (0.016)
New York 1850	0.051 (0.027)	0.011 (0.036)	0.041 (0.149)	-0.037 (0.095)	0.209** (0.070)	0.326*** (0.078)
New York 1860	0.010 (0.012)	0.144*** (0.026)	0.046 (0.085)	0.056 (0.080)	0.032 (0.043)	0.135 (0.115)
New York 1880	-0.040** (0.011)	-0.073 (0.040)	-0.164 (0.126)	-0.057 (0.065)	-0.078 (0.069)	0.041 (0.024)
New York 1900	-0.008 (0.009)	-0.056 (0.032)	-0.187** (0.056)	-0.056 (0.032)	-0.015 (0.039)	0.026 (0.016)
Ohio 1850	-0.007 (0.027)	-0.092* (0.036)	-0.014 (0.149)	-0.012 (0.095)	0.038 (0.070)	0.309*** (0.078)
Ohio 1860	0.004 (0.012)	-0.009 (0.026)	0.018 (0.085)	0.044 (0.080)	-0.115* (0.043)	0.201 (0.115)
Ohio 1880	-0.009 (0.011)	-0.124** (0.040)	-0.060 (0.126)	-0.047 (0.065)	-0.086 (0.069)	0.018 (0.024)
Ohio 1900	-0.000 (0.009)	-0.152*** (0.032)	-0.058 (0.056)	-0.015 (0.032)	-0.047 (0.039)	0.051** (0.016)
Wisconsin 1850	0.042 (0.027)	-0.058 (0.036)	0.273 (0.149)	0.068 (0.095)	0.116 (0.070)	0.273** (0.078)
Wisconsin 1860	0.103*** (0.012)	-0.005 (0.026)	-0.182* (0.085)	0.017 (0.080)	-0.240*** (0.043)	0.135 (0.115)
Wisconsin 1880	-0.071*** (0.011)	-0.103* (0.040)	-0.124 (0.126)	-0.097 (0.065)	-0.296*** (0.069)	-0.094*** (0.024)

Wisconsin 1900	-0.002 (0.009)	-0.047 (0.032)	-0.136* (0.056)	-0.127*** (0.032)	-0.110* (0.039)	-0.129*** (0.016)
<u>Cohort 1891-1900</u>						
DC 1850	-0.090*** (0.018)	0.262*** (0.054)	-0.121 (0.142)	-0.185*** (0.045)	-0.021 (0.077)	-0.027 (0.091)
DC 1860	0.021*** (0.003)	0.289*** (0.030)	0.390** (0.107)	0.121*** (0.025)	-0.434*** (0.027)	0.086 (0.098)
DC 1870	0.007 (0.011)	0.044 (0.040)	-0.060 (0.126)	-0.099 (0.065)	-0.383*** (0.069)	-0.133*** (0.024)
DC 1900	-0.016*** (0.002)	0.152** (0.050)	0.081 (0.093)	0.153*** (0.038)	-0.122* (0.053)	-0.126** (0.038)
Illinois 1850	0.053** (0.018)	0.389*** (0.054)	0.031 (0.142)	0.257*** (0.045)	0.396*** (0.077)	0.433*** (0.091)
Illinois 1860	0.051*** (0.003)	0.316*** (0.030)	0.375** (0.107)	0.191*** (0.025)	0.163*** (0.027)	0.138 (0.098)
Illinois 1870	0.007 (0.011)	0.109* (0.040)	0.042 (0.126)	-0.007 (0.065)	0.162* (0.069)	0.025 (0.024)
Illinois 1900	0.010*** (0.002)	0.049 (0.050)	0.047 (0.093)	0.025 (0.038)	0.110 (0.053)	0.065 (0.038)
Indiana 1850	-0.117*** (0.018)	0.281*** (0.054)	-0.233 (0.142)	0.129** (0.045)	0.445*** (0.077)	-0.128 (0.091)
Indiana 1860	0.078*** (0.003)	0.077* (0.030)	0.093 (0.107)	-0.112*** (0.025)	-0.166*** (0.027)	0.139 (0.098)
Indiana 1870	-0.020 (0.011)	0.088* (0.040)	0.018 (0.126)	-0.220** (0.065)	0.042 (0.069)	-0.140*** (0.024)
Indiana 1900	-0.002 (0.002)	0.119* (0.050)	-0.046 (0.093)	-0.035 (0.038)	0.164** (0.053)	0.091* (0.038)
Kentucky 1850	-0.090*** (0.018)	-0.254*** (0.054)	-0.355* (0.142)	-0.022 (0.045)	-0.029 (0.077)	0.159 (0.091)
Kentucky 1860	0.021*** (0.003)	-0.048 (0.030)	-0.111 (0.107)	0.182*** (0.025)	-0.517*** (0.027)	0.155 (0.098)
Kentucky 1870	0.007 (0.011)	-0.043 (0.040)	-0.076 (0.126)	0.056 (0.065)	0.101 (0.069)	-0.114*** (0.024)
Kentucky 1900	0.016*** (0.002)	0.197*** (0.050)	0.084 (0.093)	-0.039 (0.038)	0.024 (0.053)	0.124** (0.038)
Pennsylvania 1850	0.040* (0.018)	-0.020 (0.054)	-0.086 (0.142)	-0.209*** (0.045)	0.004 (0.077)	0.067 (0.091)
Pennsylvania 1860	0.103*** (0.003)	0.145*** (0.030)	-0.067 (0.107)	-0.017 (0.025)	-0.068* (0.027)	0.001 (0.098)
Pennsylvania 1870	-0.008 (0.011)	-0.028 (0.040)	-0.133 (0.126)	-0.149* (0.065)	0.054 (0.069)	-0.150*** (0.024)
Pennsylvania 1900	0.011*** (0.002)	-0.039 (0.050)	-0.192 (0.093)	-0.127** (0.038)	0.033 (0.053)	-0.100* (0.038)
Rhode Island 1850	0.182*** (0.018)	0.048 (0.054)	0.132 (0.142)	-0.136** (0.045)	0.088 (0.077)	0.344** (0.091)
Rhode Island 1860	0.146*** (0.003)	0.353*** (0.030)	0.302* (0.107)	0.242*** (0.025)	-0.150*** (0.027)	0.206* (0.098)
Rhode Island 1870	0.007 (0.011)	0.187*** (0.040)	-0.007 (0.126)	0.103 (0.065)	-0.270*** (0.069)	-0.165*** (0.024)
Rhode Island 1900	0.064*** (0.002)	0.031 (0.050)	-0.261* (0.093)	0.046 (0.038)	0.046 (0.053)	-0.008 (0.038)
West Virginia 1850	0.000 (.)	-0.167** (0.054)	-0.466** (0.142)	0.415*** (0.045)	-0.269** (0.077)	0.534*** (0.091)
West Virginia 1860	0.021*** (0.003)	0.000 (.)	-0.421*** (0.107)	0.507*** (0.025)	0.223*** (0.027)	0.397*** (0.098)
West Virginia 1870	0.007 (0.011)	0.044 (0.040)	-0.330* (0.126)	0.215** (0.065)	-0.006 (0.069)	0.261*** (0.024)
West Virginia 1900	-0.016*** (0.002)	-0.001 (0.050)	-0.188 (0.093)	0.146** (0.038)	0.028 (0.053)	-0.014 (0.038)
connecticut	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)
delaware	-0.203*** (0.011)	-0.386*** (0.019)	-0.545*** (0.055)	-0.216*** (0.039)	0.021 (0.040)	-0.350*** (0.009)
district of columbia	-0.188*** (0.011)	-0.544*** (0.040)	-0.481** (0.126)	-0.088 (0.065)	-0.042 (0.069)	-0.075** (0.024)
idaho	-0.204*** (0.009)			-0.735*** (0.032)	0.186*** (0.039)	0.183*** (0.016)
illinois	-0.188*** (0.011)	-0.499*** (0.040)	-0.333* (0.126)	-0.030 (0.065)	-0.126 (0.069)	-0.035 (0.024)
indiana	-0.162*** (0.011)	-0.491*** (0.040)	-0.236 (0.126)	0.098 (0.065)	-0.175* (0.069)	-0.045 (0.024)
iowa	-0.144*** (0.007)	-0.298*** (0.020)	-0.085 (0.060)	0.085* (0.036)	0.018 (0.036)	-0.040*** (0.005)
kentucky	-0.188*** (0.011)	-0.457*** (0.040)	-0.447** (0.126)	0.026 (0.065)	-0.101 (0.069)	-0.094*** (0.024)

maine	0.068*** (0.000)	0.038*** (0.000)	0.233*** (0.000)	0.146*** (0.000)	0.200*** (0.000)	0.125*** (0.000)
maryland	-0.167*** (0.010)	-0.412*** (0.020)	-0.407*** (0.061)	-0.095 (0.046)	-0.037 (0.038)	0.057** (0.016)
massachusetts	-0.096*** (0.000)	-0.079*** (0.000)	0.174*** (0.000)	0.238*** (0.000)	0.156*** (0.000)	0.051*** (0.000)
michigan	-0.126*** (0.000)	-0.333*** (0.000)	-0.048*** (0.000)	0.040*** (0.000)	0.082*** (0.000)	-0.113*** (0.000)
missouri	-0.174*** (0.008)	-0.466*** (0.019)	-0.360*** (0.059)	-0.001 (0.036)	-0.019 (0.034)	0.012 (0.006)
new hampshire	-0.182*** (0.000)	-0.100*** (0.000)	-0.417*** (0.000)	-0.188*** (0.000)	-0.133*** (0.000)	0.125*** (0.000)
new jersey	-0.146*** (0.000)	-0.380*** (0.000)	-0.121*** (0.000)	-0.013*** (0.000)	0.129*** (0.000)	-0.014*** (0.000)
new york	-0.128*** (0.000)	-0.279*** (0.000)	-0.081*** (0.000)	0.087*** (0.000)	-0.036*** (0.000)	-0.051*** (0.000)
ohio	-0.172*** (0.000)	-0.333*** (0.000)	-0.180*** (0.000)	0.047*** (0.000)	0.044*** (0.000)	0.018*** (0.000)
pennsylvania	-0.162*** (0.011)	-0.375*** (0.040)	-0.113 (0.126)	0.115 (0.065)	-0.030 (0.069)	0.103*** (0.024)
rhode island	-0.188*** (0.011)	-0.258*** (0.040)	-0.060 (0.126)	0.077 (0.065)	0.015 (0.069)	0.054* (0.024)
west virginia	-0.188*** (0.011)	-0.544*** (0.040)	-0.337* (0.126)	-0.188** (0.065)	-0.128 (0.069)	-0.136*** (0.024)
wisconsin	-0.071*** (0.000)	-0.152*** (0.000)	-0.076*** (0.000)	0.159*** (0.000)	0.155*** (0.000)	0.125*** (0.000)
T=1860	-0.111*** (0.018)	-0.205*** (0.036)	-0.044 (0.204)	-0.092 (0.047)	0.174* (0.073)	0.138 (0.150)
T=1870	-0.097** (0.027)	-0.211*** (0.036)	-0.136 (0.149)	-0.086 (0.095)	0.070 (0.070)	0.273** (0.078)
T=1880	-0.090*** (0.018)	-0.167** (0.054)	-0.091 (0.142)	-0.085 (0.045)	0.112 (0.077)	0.201* (0.091)
T=1900	-0.074*** (0.019)	-0.165*** (0.028)	-0.027 (0.170)	-0.039 (0.067)	0.084* (0.034)	0.215** (0.071)
Constant	0.279*** (0.027)	0.711*** (0.036)	0.803*** (0.149)	0.773*** (0.095)	0.730*** (0.070)	0.602*** (0.078)
<u>Pre-trends</u>						
Exposure T-4	0.032 (0.029)	0.043 (0.077)	-0.096 (0.146)	-0.121 (0.081)	0.036 (0.086)	0.145 (0.116)
Exposure T-3	0.099*** (0.033)	0.109** (0.047)	0.007 (0.055)	0.001 (0.059)	0.008 (0.061)	0.184* (0.090)
Exposure T-2	0.021 (0.014)	0.085 ** (0.032)	-0.037 (0.061)	-0.007 (0.043)	-0.006 (0.037)	0.023 (0.073)
R2	0.075	0.120	0.073	0.030	0.038	0.030
N	5672.000	5346.000	5264.000	5059.000	4881.000	4584.000

Appendix Table 2. Effect of compulsory schooling on (linear) probability of being in school, by age (ages 10-16)

	(1) OLS: Age 10	(2) OLS: Age 11	(3) OLS: Age 12	(4) OLS: Age 13	(5) OLS: Age 14	(6) OLS: Age 15	(7) OLS: Age 16
Cohort 1871-1880							
Connecticut 1850	0.186*** (0.021)	0.195 (0.141)	0.268** (0.075)	0.323*** (0.041)	-0.015 (0.098)	-0.240 (0.174)	-0.251*** (0.050)
Connecticut 1860	0.058 (0.089)	-0.052 (0.048)	0.002 (0.111)	0.367*** (0.065)	-0.007 (0.066)	0.136 (0.142)	0.252*** (0.061)
Connecticut 1880	-0.126* (0.052)	0.007 (0.089)	-0.159*** (0.033)	0.148** (0.040)	0.224 (0.130)	-0.277*** (0.068)	0.116* (0.048)
Connecticut 1900	-0.076** (0.026)	0.011 (0.041)	-0.038 (0.023)	0.268*** (0.040)	0.142* (0.060)	-0.195 (0.107)	0.154*** (0.032)
Maine 1850	0.203*** (0.021)	0.108 (0.141)	0.268** (0.075)	0.326*** (0.041)	0.063 (0.098)	-0.252 (0.174)	0.502*** (0.050)
Maine 1860	0.111 (0.089)	-0.180** (0.048)	0.002 (0.111)	0.426*** (0.065)	0.099 (0.066)	0.040 (0.142)	0.505*** (0.061)
Maine 1880	0.009 (0.052)	-0.144 (0.089)	0.041 (0.033)	0.484*** (0.040)	-0.061 (0.130)	-0.222** (0.068)	-0.087 (0.048)
Maine 1900	0.092** (0.026)	-0.050 (0.041)	-0.013 (0.023)	0.347*** (0.040)	-0.030 (0.060)	-0.288* (0.107)	-0.096** (0.032)
Massachusetts 1850	0.208*** (0.021)	0.137 (0.141)	0.301*** (0.075)	0.091* (0.041)	0.018 (0.098)	-0.127 (0.174)	0.076 (0.050)
Massachusetts 1860	0.067 (0.089)	0.003 (0.048)	0.169 (0.111)	0.092 (0.065)	0.231** (0.066)	0.010 (0.142)	0.151* (0.061)
Massachusetts 1880	0.068 (0.052)	0.064 (0.089)	-0.008 (0.033)	0.059 (0.040)	0.034 (0.130)	0.051 (0.068)	0.083 (0.048)
Massachusetts 1900	-0.051 (0.026)	-0.008 (0.041)	0.064* (0.023)	0.061 (0.040)	0.023 (0.060)	-0.030 (0.107)	0.118** (0.032)
Michigan 1850	0.311*** (0.021)	0.371* (0.141)	0.304*** (0.075)	0.353*** (0.041)	0.129 (0.098)	-0.863*** (0.174)	0.000 (.)
Michigan 1860	0.260** (0.089)	0.139** (0.048)	0.204 (0.111)	0.397*** (0.065)	0.082 (0.066)	-0.209 (0.142)	0.467*** (0.061)
Michigan 1880	0.000 (0.052)	0.222* (0.089)	-0.240*** (0.033)	0.122** (0.040)	-0.054 (0.130)	-0.148* (0.068)	0.008 (0.048)
Michigan 1900	0.004 (0.026)	0.153** (0.041)	-0.050* (0.023)	0.149** (0.040)	-0.092 (0.060)	0.052 (0.107)	0.090* (0.032)
New Hampshire 1850	0.269*** (0.021)	0.051 (0.141)	0.697*** (0.075)	0.217*** (0.041)	0.068 (0.098)	0.248 (0.174)	-0.037 (0.050)
New Hampshire 1860	0.219* (0.089)	0.129* (0.048)	0.168 (0.111)	0.262*** (0.065)	-0.157* (0.066)	-0.126 (0.142)	0.383*** (0.061)
New Hampshire 1880	0.058 (0.052)	0.181 (0.089)	0.469*** (0.033)	-0.291*** (0.040)	-0.137 (0.130)	0.178* (0.068)	-0.087 (0.048)
New Hampshire 1900	0.032 (0.026)	0.093* (0.041)	0.416*** (0.023)	-0.004 (0.040)	-0.219** (0.060)	-0.326** (0.107)	-0.498*** (0.032)
New Jersey 1850	0.323*** (0.021)	0.153 (0.141)	0.365*** (0.075)	0.211*** (0.041)	0.006 (0.098)	-0.294 (0.174)	-0.185** (0.050)
New Jersey 1860	0.172 (0.089)	0.081 (0.048)	0.210 (0.111)	0.121 (0.065)	0.129 (0.066)	0.325* (0.142)	0.271*** (0.061)
New Jersey 1880	0.101 (0.052)	0.023 (0.089)	0.018 (0.033)	-0.030 (0.040)	0.056 (0.130)	0.046 (0.068)	0.095 (0.048)
New Jersey 1900	0.038 (0.026)	0.099* (0.041)	0.089*** (0.023)	0.184*** (0.040)	-0.009 (0.060)	0.080 (0.107)	0.073* (0.032)
New York 1850	0.208*** (0.021)	0.200 (0.141)	0.284** (0.075)	0.220*** (0.041)	-0.200 (0.098)	-0.475* (0.174)	-0.090 (0.050)
New York 1860	0.069 (0.089)	-0.013 (0.048)	0.234* (0.111)	0.180* (0.065)	0.194** (0.066)	-0.059 (0.142)	0.015 (0.061)
New York 1880	-0.034 (0.052)	-0.039 (0.089)	-0.086* (0.033)	0.006 (0.040)	0.031 (0.130)	-0.059 (0.068)	-0.052 (0.048)
New York 1900	-0.028 (0.026)	-0.028 (0.041)	0.055* (0.023)	0.167*** (0.040)	0.064 (0.060)	0.058 (0.107)	-0.029 (0.032)
Ohio 1850	0.183*** (0.021)	0.225 (0.141)	0.201* (0.075)	0.178*** (0.041)	-0.380*** (0.098)	-0.485* (0.174)	-0.309*** (0.050)
Ohio 1860	0.077 (0.089)	-0.047 (0.048)	0.129 (0.111)	0.128 (0.065)	-0.112 (0.066)	-0.123 (0.142)	0.158* (0.061)
Ohio 1880	-0.030 (0.052)	0.018 (0.089)	-0.126** (0.033)	-0.069 (0.040)	-0.223 (0.130)	-0.169* (0.068)	-0.047 (0.048)
Ohio 1900	-0.007 (0.026)	0.055 (0.041)	0.006 (0.023)	0.066 (0.040)	-0.049 (0.060)	-0.195 (0.107)	-0.109** (0.032)
Wisconsin 1850	0.217*** (0.021)	0.208 (0.141)	0.030 (0.075)	0.000 (.)	0.000 (.)	0.498** (0.174)	0.000 (.)
Wisconsin 1860	0.166 (0.089)	-0.247*** (0.048)	0.121 (0.111)	0.004 (0.065)	0.276*** (0.066)	0.224 (0.142)	0.479*** (0.061)
Wisconsin 1880	-0.173** (0.052)	-0.012 (0.089)	-0.125** (0.033)	-0.058 (0.040)	0.007 (0.130)	0.190* (0.068)	0.176** (0.048)

Wisconsin 1900	-0.051 (0.026)	0.020 (0.041)	0.059* (0.023)	-0.054 (0.040)	0.171** (0.060)	0.151 (0.107)	0.181*** (0.032)
<u>Cohort 1881-1900</u>							
DC 1850	0.350*** (0.034)	0.177* (0.064)	0.000 (.)	0.067 (0.064)	-0.264*** (0.040)	0.010 (0.122)	-0.617*** (0.075)
DC 1860	-0.033 (0.048)	0.074 (0.073)	0.294** (0.091)	0.112 (0.061)	0.411*** (0.071)	-0.098 (0.076)	0.220*** (0.029)
DC 1870	-0.085 (0.052)	0.112 (0.089)	0.126** (0.033)	-0.025 (0.040)	0.334* (0.130)	-0.238** (0.068)	0.253*** (0.048)
DC 1900	0.114** (0.033)	0.063 (0.056)	-0.029 (0.021)	-0.202*** (0.030)	0.139 (0.070)	-0.310*** (0.048)	-0.203*** (0.046)
Illinois 1850	-0.353*** (0.034)	0.326*** (0.064)	0.322*** (0.050)	0.354*** (0.064)	0.103* (0.040)	0.227 (0.122)	0.277** (0.075)
Illinois 1860	0.054 (0.048)	0.061 (0.073)	0.199* (0.091)	0.398*** (0.061)	0.402*** (0.071)	0.253** (0.076)	0.417*** (0.029)
Illinois 1870	-0.067 (0.052)	0.063 (0.089)	0.086* (0.033)	-0.034 (0.040)	0.159 (0.130)	0.133 (0.068)	-0.067 (0.048)
Illinois 1900	-0.069 (0.033)	0.084 (0.056)	0.119*** (0.021)	0.162*** (0.030)	0.030 (0.070)	-0.084 (0.048)	-0.059 (0.046)
Indiana 1850	0.033 (0.034)	0.237** (0.064)	0.172** (0.050)	-0.059 (0.064)	0.331*** (0.040)	0.194 (0.122)	-0.026 (0.075)
Indiana 1860	0.010 (0.048)	-0.085 (0.073)	0.211* (0.091)	0.235** (0.061)	-0.161* (0.071)	-0.180* (0.076)	0.228*** (0.029)
Indiana 1870	0.033 (0.052)	-0.175 (0.089)	-0.020 (0.033)	0.121** (0.040)	0.202 (0.130)	0.160* (0.068)	-0.007 (0.048)
Indiana 1900	0.021 (0.033)	0.027 (0.056)	0.072** (0.021)	0.355*** (0.030)	0.101 (0.070)	-0.071 (0.048)	0.121* (0.046)
Kentucky 1850	0.185*** (0.034)	0.270*** (0.064)	0.401*** (0.050)	-0.695*** (0.064)	0.070 (0.040)	-0.127 (0.122)	-0.331*** (0.075)
Kentucky 1860	-0.282*** (0.048)	0.058 (0.073)	0.243* (0.091)	0.149* (0.061)	0.203** (0.071)	0.099 (0.076)	0.061* (0.029)
Kentucky 1870	0.250*** (0.052)	0.019 (0.089)	0.311*** (0.033)	-0.010 (0.040)	-0.191 (0.130)	0.037 (0.068)	0.081 (0.048)
Kentucky 1900	0.129** (0.033)	0.027 (0.056)	0.179*** (0.021)	0.043 (0.030)	0.054 (0.070)	0.079 (0.048)	0.008 (0.046)
Pennsylvania 1850	0.040 (0.034)	0.051 (0.064)	0.186** (0.050)	-0.145* (0.064)	-0.067 (0.040)	-0.242 (0.122)	-0.034 (0.075)
Pennsylvania 1860	0.032 (0.048)	-0.045 (0.073)	0.121 (0.091)	0.100 (0.061)	0.229** (0.071)	0.126 (0.076)	0.170*** (0.029)
Pennsylvania 1870	-0.106 (0.052)	-0.025 (0.089)	0.012 (0.033)	-0.011 (0.040)	0.048 (0.130)	0.042 (0.068)	-0.053 (0.048)
Pennsylvania 1900	-0.073* (0.033)	-0.022 (0.056)	0.068** (0.021)	-0.085* (0.030)	-0.015 (0.070)	-0.034 (0.048)	-0.026 (0.046)
Rhode Island 1850	0.397*** (0.034)	-0.087 (0.064)	0.278*** (0.050)	0.175* (0.064)	-0.243*** (0.040)	-1.032*** (0.122)	-0.083 (0.075)
Rhode Island 1860	0.347*** (0.048)	0.034 (0.073)	0.294** (0.091)	0.086 (0.061)	0.003 (0.071)	-0.140 (0.076)	0.468*** (0.029)
Rhode Island 1870	0.129* (0.052)	0.028 (0.089)	0.147*** (0.033)	0.249*** (0.040)	0.069 (0.130)	-0.405*** (0.068)	0.165** (0.048)
Rhode Island 1900	0.161*** (0.033)	0.089 (0.056)	0.086*** (0.021)	0.182*** (0.030)	0.030 (0.070)	-0.606*** (0.048)	-0.196*** (0.046)
West Virginia 1850	-0.630*** (0.034)	0.320*** (0.064)	0.000 (.)	0.000 (.)	0.206*** (0.040)	-0.692*** (0.122)	-0.450*** (0.075)
West Virginia 1860	0.319*** (0.048)	0.199* (0.073)	0.000 (.)	0.553*** (0.061)	0.380*** (0.071)	0.034 (0.076)	0.387*** (0.029)
West Virginia 1870	0.041 (0.052)	0.212* (0.089)	0.293*** (0.033)	0.416*** (0.040)	0.054 (0.130)	0.060 (0.068)	-0.080 (0.048)
West Virginia 1900	-0.034 (0.033)	0.163** (0.056)	0.280*** (0.021)	0.287*** (0.030)	-0.082 (0.070)	-0.180** (0.048)	-0.211*** (0.046)
connecticut	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)
delaware	-0.189*** (0.033)	-0.150** (0.042)	-0.348*** (0.022)	-0.210*** (0.025)	0.086 (0.101)	-0.203* (0.082)	0.075* (0.031)
district of columbia	-0.165** (0.052)	-0.125 (0.089)	-0.126** (0.033)	0.256*** (0.040)	0.083 (0.130)	0.150* (0.068)	0.366*** (0.048)
illinois	-0.062 (0.052)	-0.131 (0.089)	-0.197*** (0.033)	-0.031 (0.040)	-0.034 (0.130)	-0.068 (0.068)	0.138** (0.048)
indiana	-0.097 (0.052)	-0.042 (0.089)	-0.154*** (0.033)	-0.118** (0.040)	-0.012 (0.130)	-0.034 (0.068)	0.108* (0.048)
iowa	-0.121*** (0.029)	-0.032 (0.043)	-0.076** (0.026)	0.156*** (0.030)	0.160* (0.056)	0.034 (0.081)	0.251*** (0.029)
kentucky	-0.250*** (0.052)	-0.075 (0.089)	-0.383*** (0.033)	0.018 (0.040)	-0.085 (0.130)	-0.213** (0.068)	0.080 (0.048)
maine	-0.143*** (0.000)	0.087*** (0.000)	0.000 (0.000)	-0.169*** (0.000)	0.144*** (0.000)	0.245*** (0.000)	0.247*** (0.000)

maryland	-0.050 (0.033)	-0.018 (0.047)	-0.117*** (0.030)	-0.004 (0.029)	-0.032 (0.064)	-0.280** (0.082)	0.080* (0.029)
massachusetts	-0.052*** (0.000)	0.005*** (0.000)	-0.106*** (0.000)	0.132*** (0.000)	0.106*** (0.000)	-0.047*** (0.000)	0.092*** (0.000)
michigan	-0.125*** (0.000)	-0.176*** (0.000)	-0.036*** (0.000)	-0.030*** (0.000)	0.189*** (0.000)	0.023*** (0.000)	0.119*** (0.000)
missouri	-0.107** (0.029)	-0.030 (0.042)	-0.088** (0.025)	0.076* (0.030)	0.034 (0.066)	-0.119 (0.082)	0.113*** (0.026)
new hampshire	-0.083*** (0.000)	-0.056*** (0.000)	-0.429*** (0.000)	0.106*** (0.000)	0.250*** (0.000)	-0.088*** (0.000)	0.286*** (0.000)
new jersey	-0.137*** (0.000)	-0.125*** (0.000)	-0.208*** (0.000)	-0.087*** (0.000)	-0.021*** (0.000)	-0.297*** (0.000)	-0.066*** (0.000)
new york	-0.104*** (0.000)	-0.032*** (0.000)	-0.159*** (0.000)	-0.043*** (0.000)	-0.067*** (0.000)	-0.207*** (0.000)	0.097*** (0.000)
ohio	-0.084*** (0.000)	-0.030*** (0.000)	-0.076*** (0.000)	0.054*** (0.000)	0.198*** (0.000)	0.008*** (0.000)	0.201*** (0.000)
pennsylvania	-0.055 (0.052)	-0.023 (0.089)	-0.212*** (0.033)	0.048 (0.040)	-0.085 (0.130)	-0.209** (0.068)	0.094 (0.048)
rhode island	-0.212*** (0.052)	-0.052 (0.089)	-0.209*** (0.033)	-0.019 (0.040)	0.062 (0.130)	0.192* (0.068)	0.166** (0.048)
west virginia	-0.184** (0.052)	-0.125 (0.089)	-0.293*** (0.033)	-0.185*** (0.040)	0.113 (0.130)	-0.148* (0.068)	0.199*** (0.048)
wisconsin	-0.031*** (0.000)	-0.013*** (0.000)	-0.095*** (0.000)	0.113*** (0.000)	-0.117*** (0.000)	-0.338*** (0.000)	-0.060*** (0.000)
idaho			-0.013 (0.023)		0.364*** (0.060)	-0.114 (0.107)	0.788*** (0.032)
T=1860	0.050 (0.077)	0.121 (0.128)	0.100 (0.074)	-0.044 (0.041)	-0.175** (0.054)	-0.226 (0.109)	-0.170 (0.087)
T=1870	0.186*** (0.021)	0.108 (0.141)	0.268** (0.075)	0.092* (0.041)	-0.098 (0.098)	-0.252 (0.174)	-0.037 (0.050)
T=1880	0.120** (0.034)	0.070 (0.064)	0.228*** (0.050)	0.008 (0.064)	-0.128** (0.040)	-0.263* (0.122)	-0.117 (0.075)
T=1900	0.237*** (0.012)	0.158 (0.101)	0.281*** (0.061)	0.055 (0.047)	-0.046 (0.040)	-0.226* (0.103)	-0.038 (0.061)
Constant	0.814*** (0.021)	0.805*** (0.141)	0.732*** (0.075)	0.677*** (0.041)	0.682*** (0.098)	0.840*** (0.174)	0.251*** (0.050)
<u>Pre-trends</u>							
Exposure T-4	0.058 (0.058)	0.101 (0.076)	0.221*** (0.058)	-0.091 (0.090)	-0.054 (0.052)	-0.215 (0.137)	-0.065 (0.087)
Exposure T-3	0.132** (0.051)	0.077 (0.061)	0.231** (0.085)	0.167** (0.060)	0.044 (0.086)	-0.149 (0.172)	0.085 (0.064)
Exposure T-2	0.019 (0.044)	-0.006 (0.043)	0.115* (0.059)	0.083* (0.048)	0.105* (0.055)	0.011 (0.048)	0.061 (0.044)
R2	0.049	0.038	0.062	0.061	0.051	0.058	0.043
N	4743.000	4080.000	4354.000	3713.000	3934.000	3458.000	3374.000